Nasrul Huda

Data Scientist

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Photo
(3.5 × 4.5 cm)

About Me

Data Science and AI student with hands-on experience in machine learning, full-stack development, and research. Passionate about building intelligent systems that solve real-world problems. Strong foundation in Python, modern web technologies, and cloud infrastructure. Seeking opportunities to apply AI/ML expertise in innovative projects.

Professional Experience

SynTwin GmbH — Munich, Germany
AI Engineer (Working Student) — *October 2025 – April 2026*

- Developed agentic AI workflows and orchestration pipelines for a Digital Twin platform, designing multi-turn conversational agent architectures using Python and LiveKit.
- Applied prompt engineering strategies to improve agent reliability and consistency across complex, stateful interactions.
- Built and maintained React/Vite frontend components integrated with Supabase for real-time data sync and user authentication.
- Collaborated closely with the CTO and a team of 5 developers on end-to-end system design spanning AI agent architecture, backend services, and user-facing interfaces.
- **Tech:** LiveKit, Supabase, React, Docker Compose, STT, TTS, GitHub Actions

Universität Hamburg – Security & Distributed Systems — Hamburg, Germany
Research Assistant — *July 2025 – September 2025*

- Built Python-based microservices for signing and verifying cryptographic hash signatures as part of research data management and security analysis tooling.
- Contributed to the NFDIxCS (National Research Data Infrastructure for Computer Science) project, working at the intersection of AI, data science, and distributed systems.
- **Tech:** Flask, ECDSA, RSA, Docker, GitLab CI, Linux, Cybersecurity, Distributed Systems

Pettoo UG — Hamburg, Germany
Freelance Software Engineer — *June 2025 – September 2025*

- Designed and implemented the backend of a pet services platform enabling users to discover and book services, purchase products, and connect with other pet owners, using FastAPI and PostgreSQL.
- Architected the database schema and core business logic, and integrated Elasticsearch for search functionality.
- Led a team of 3 developers, coordinating tasks and maintaining consistent code quality across the project.
- Set up CI/CD pipelines with GitHub Actions and deployed containerised services on Google Cloud Platform.
- **Tech:** FastAPI, PostgreSQL, Google Cloud Platform, Docker, Elasticsearch, GitHub Actions

Universität Hamburg – Language Technology Research Group — Hamburg, Germany
Research Assistant — *February 2025–May 2025*

- Contributed to DATS Project (Discourse Analysis Tool Suite), an NLP platform for automated text analysis.
- Developed full-stack components: React TypeScript frontend, FastAPI backend with RESTful APIs.
- Implemented NLP processing modules and text preprocessing pipelines for large-scale linguistic datasets.
- **Tech:** FastAPI, React, NLP, Machine Learning, Docker, Git

Maseehum Task Manager New Delhi, India
Software Developer *April 2023–October 2023*

- Developed and maintained REST APIs using FastAPI for order processing, validation, and transaction workflows, with database schema design and migrations on Supabase (PostgreSQL) for scalable data management.
- Built a Wix order synchronisation module for automated customer order fetching, integrated Zoho Books API for invoice generation and financial workflow automation, and WhatsApp API for automated notifications.
- Developed a real-time messaging system using WebSockets to improve system responsiveness and user interaction, and collaborated with the team to debug issues, optimise performance, and enhance system reliability.
- **Tech:** FastAPI, Supabase, PostgreSQL, WebSockets, Zoho Books API, WhatsApp API

Education

Universität Hamburg Hamburg, Germany
M.Sc. in Data Science and Artificial Intelligence *October 2024–March 2027*
Advanced study of machine learning, deep learning, statistical modelling, and AI systems, with emphasis on data-driven decision making, data privacy, natural language processing, computer vision, and large-scale data engineering.

Jamia Millia Islamia New Delhi, India
B.Tech in Computer Science Engineering *October 2020–June 2024*
Comprehensive study of computer science including algorithms, data structures, operating systems, database management, software engineering, computer networks, cryptography, computer architecture and object-oriented programming.

Laxman Public School New Delhi, India
Higher Secondary Education (CBSE) *2020*
Field of Study: Physics, Chemistry, Mathematics, Computer Science

The Frank Anthony Public School New Delhi, India
Secondary Education (ICSE) *2018*
Field of Study: Science

Projects

Few-Shot Bioacoustic Sound Event Detection using Prototypical Networks
Torchaudio, PyTorch, PyTorch Lightning, Hydra, MLflow

- Developed a few-shot meta-learning system to classify animal vocalizations from as few as 5 labeled examples per species, using episodic (N-way, K-shot) training with Prototypical Networks.
- Engineered audio features from raw .wav files into 128-bin log-mel spectrograms and PCEN arrays using librosa, pre-computed and stored as .npy for efficient training.
- Designed and compared four encoder architectures: ResNet baseline (V1), ResNet with channel & temporal attention + learnable distance metric (V2), Audio Spectrogram Transformer with multi-head self-attention (V3), and pretrained EfficientNet-B1 (V4).
- Built end-to-end training infrastructure with PyTorch Lightning, Hydra configuration management, MLflow experiment tracking, and a custom CLI for training, evaluation, and feature export.

Federated Fine-Tuning of Nicheformer
PyTorch, Flower, Federated Learning

- Implemented federated learning framework for fine-tuning Nicheformer transformer model on mouse brain spatial transcriptomics data, comparing centralized (94.45%), federated (92.76%), and local training strategies.
- Built data pipeline for anatomical non-IID partitioning (dorsal/mid/ventral regions) with 24-class cell type classification on 250-dimensional features (248 genes + spatial coordinates) achieving competitive 92.76% accuracy in federated setting.

ACL Anthology Semantic Retrieval System
FastAPI, Langchain, React, Qdrant, Fireworks AI, Groq

- Built a Retrieval-Augmented Generation (RAG) system over the ACL Anthology corpus to enable semantic search across tens of thousands of NLP research papers, going beyond keyword matching to understand query meaning using dense vector embeddings.
- Implemented an offline ingestion pipeline that fetches paper metadata and abstracts, generates dense embeddings via Fireworks AI, and indexes them into a Qdrant vector store for fast approximate nearest-neighbour retrieval.

- Designed two query modes: natural language input (reformulated into multiple semantic queries by an LLM via Groq) and "Paper as Query" (using a paper's abstract as a semantic proxy to find related work); results aggregated using Reciprocal Rank Fusion (RRF).
- Developed a FastAPI backend and React frontend with real-time streaming responses, inline citations with hover previews, and a query monitoring panel showing pipeline steps and timing.

MyTorch: PyTorch-Inspired Deep Learning Framework
NumPy, Deep Learning

- Built a minimal deep learning library from scratch inspired by PyTorch's API, implementing core tensor operations (arithmetic, matrix multiplication, transpose) with a custom automatic differentiation engine and backward pass for gradient computation.
- Implemented neural network modules including fully connected (Linear) layers, ReLU activations, and Cross Entropy loss, all with manual forward and backward propagation with no ML framework dependencies.
- Developed SGD and Adam optimisers from first principles, replicating PyTorch's modular optimizer interface.
- Validated the framework on MNIST digit classification, achieving **97.16% test accuracy** (macro F1: 97.15%) using only linear layers, demonstrating correctness of the custom autograd and training pipeline.

GitGood: AI-Powered GitHub Repository Analysis Tool
FastAPI, React, Docker, OpenAI API

- Built a full-stack web application that integrates with GitHub OAuth to let users import repositories and interact with their codebase through an AI-powered chat interface.
- Designed a FastAPI backend with SQLAlchemy for user session management and repository data storage, connected to the OpenAI API to answer natural language questions about commit history, diffs, and code evolution.
- Developed a responsive React/TypeScript frontend with Vite and TailwindCSS, with Axios handling API communication between client and server.
- Containerised the full stack using Docker and Docker Compose for consistent local development and production deployment.

Technical Skills

Languages:	Python, TypeScript, C++, SQL, Rust(Novice)
Frameworks:	PyTorch, PyTorch Lightning, Scikit-Learn, Pandas, FastAPI, React, LangChain, Docker, Git
Databases:	PostgreSQL, Qdrant, Supabase, Redis, Elasticsearch
Cloud/DevOps:	Google Cloud Platform, Docker, CI/CD, GitHub Actions

Languages

English: C1 (Advanced) **Hindi:** Native **Urdu:** Native **German:** A2 (Elementary)

Awards & Achievements

- **Letter of Excellence** – Vice Chancellor of Jamia Millia Islamia (outstanding academic performance).
- **IELTS 8.0 Band (C1)** – R: 9.0, L: 8.0, W: 7.5, S: 7.0.
- **1st Rank** – Top position in 4th year of Bachelor's program (2024).
- **Certificate of Excellence** – Coding Ninjas (Data Structures and Algorithms).